IDS Group Project Data Setting & Cleaning

K9

2025-12-03

Assignment brief

- Perform exploratory data analysis to assess how 6 continents in the world (all continents except Antarctica) are faring with respect to the following specific targets set out by the UN:

1. Sustain per capita economic growth in accordance with national circumstances and, in particular, at least 7 per cent gross domestic product growth per annum in the least developed countries.
2. By 2020, substantially reduce the proportion of youth not in employment, education or training.

- Find another CSV file to add to analysis

(Please note that hereafter education, employment, and training would be referred to as EET)

Data Used in the Report

In the first chunk, we would import the following data:

1. The three data sets given in the Hub;
2. The additional list of the least developed countries in the world;
3. The population of each country for each year in the interval 1960 - 2024 (World Bank, 2025a);
4. The youth unemployment rate (as a percentage of the total labour force) for each country (World Bank, 2025b);
5. The education attainment rate of the population aged 25+ for at least completing an upper secondary education (World Bank, 2025c).

* *Youth generally would be referred to the individual between age 15-24*

```r
### Please note that if you have directly cloned the GitHub repository, then you
### can just leave the code as that, **NO need to change working directory**

### And if you would like to add additional files, please just use the path starting
### with `data_sets` (as below), as ideally the default wd is the IDS_Group_Proj already

continents <- read.csv("data_sets/continents-according-to-our-world-in-data.csv")
gdp_per_capita_df <- read.csv("data_sets/gdp-per-capita-worldbank.csv")
youth_not_in_eet <- read.csv("data_sets/youth-not-in-education-employment-training.csv")

# We now create a vector of all LDC countries
ldc_countries <- c(
  "Afghanistan", "Angola", "Bangladesh", "Benin", "Bhutan", "Burkina Faso",
  "Burundi", "Cambodia", "Central African Republic", "Chad", "Comoros",
  "Democratic Republic of the Congo", "Djibouti", "Eritrea", "Ethiopia",
```

```r
  "Gambia", "Guinea", "Guinea-Bissau", "Haiti", "Kiribati",
  "Lao People's Democratic Republic", "Lesotho", "Liberia", "Madagascar",
  "Malawi", "Mali", "Mauritania", "Mozambique", "Myanmar", "Nepal",
  "Niger", "Rwanda", "Sao Tome and Principe", "Senegal", "Sierra Leone",
  "Solomon Islands", "Somalia", "South Sudan", "Sudan", "Timor-Leste",
  "Togo", "Tuvalu", "Uganda", "United Republic of Tanzania",
  "Vanuatu", "Yemen", "Zambia"
)

# Importing the population data
population <- read_csv("data_sets/population.csv", skip = 3)

# Importing the youth unemployment data
youth_unemployment <- read_csv("data_sets/youth_unemployment.csv",
  skip = 3
)


# Importing the education Attainment data
edu_attainment <- read_csv("data_sets/edu_attainment_completed_upper_secondary.csv",
  skip = 3
)
```

Original Data Sets Given

We now clean the data, getting rid of any irrelevant data.

```r
# The year is irrelevant in continents, as the only thing this df needs to be
# used for is assigning a continent to a country
continents$Year <- NULL
continents$Entity <- NULL

# We now assign the continent to each country in gdp per capita
gdp_per_capita_df <- gdp_per_capita_df %>%
  left_join(continents, by = "Code")

# Same thing for youth not in eet
youth_not_in_eet <- youth_not_in_eet %>%
  left_join(continents, by = "Code")

# The continents table is now redundant (Commented this out as when running
# this code seperately without the previous chunk it would produce error)

# rm(continents)

# We clean population data

population <- population %>%
  # Remove indicator code&name, and the last column which is all NAs
  select(-`Indicator Code`, -`Indicator Name`, -70) %>%
  rename(
    Entity = `Country Name`,
    Code = `Country Code`
  ) %>%
```

```r
  pivot_longer(
    cols = 3:67,
    names_to = "Year",
    values_to = "Population"
  )
```

Renaming columns to nicer names

```r
# Rename a column in gdp per capita
gdp_per_capita_df <- gdp_per_capita_df %>%
  rename(Gdp_per_capita = GDP.per.capita..PPP..constant.2017.international...)

# Rename a column in youth not in eet
youth_not_in_eet <- youth_not_in_eet %>%
  rename(
    Percentage_NEET =
      Share.of.youth.not.in.education..employment.or.training..total....of.youth.population.
  )
```

Calculating Economic Growth and change in eet of each country

```r
gdp_per_capita_df <- gdp_per_capita_df %>%
  mutate(Growth = ifelse(lag(Entity) != Entity, NA, (Gdp_per_capita - lag(Gdp_per_capita)) /
    lag(Gdp_per_capita) * 100))

youth_not_in_eet <- youth_not_in_eet %>%
  mutate(Perc_Change_NEET = ifelse(lag(Entity) != Entity, NA, (Percentage_NEET -
    lag(Percentage_NEET)) / lag(Percentage_NEET) * 100))
```

This chunk is now to add a flag to the gdp per capita dataframe, labelling if a country is an LDC or not

```r
gdp_per_capita_df <- gdp_per_capita_df %>%
  mutate(Ldc = Entity %in% ldc_countries)
```

Merging the data frames to include population in both

```r
gdp_per_capita_df <- merge(gdp_per_capita_df, population, by = c("Entity", "Year", "Code"))
youth_not_in_eet <- merge(youth_not_in_eet, population, by = c("Entity", "Year", "Code"))
```

Now we convert to numeric and calculate GDP per capita of each country each year

```r
gdp_per_capita_df$Population <- as.numeric(gdp_per_capita_df$Population)

gdp_per_capita_df <- gdp_per_capita_df %>%
  mutate(Gdp = Population * Gdp_per_capita)
```

Finally, export the CSV files of the 2 data frames

```r
write.csv(gdp_per_capita_df, "Exported_data/gdp_per_capita_cleaned.csv", row.names = FALSE)
write.csv(youth_not_in_eet, "Exported_data/youth_neet_cleaned.csv", row.names = FALSE)
```

Additional Data Sets

We now clean the youth unemployment df and the education attainment df.

```r
youth_unemployment <- youth_unemployment %>%
  # Remove indicator code&name, and the last column which is all NAs
  select(-`Indicator Code`, -`Indicator Name`, -70) %>%
```

```r
  rename(
    Entity = `Country Name`,
    Code = `Country Code`
  ) %>%
  pivot_longer(
    cols = 3:67,
    names_to = "Year",
    values_to = "Youth_Unemployment"
  )

youth_unemployment$Year <- as.numeric(youth_unemployment$Year)

edu_attainment <- edu_attainment %>%
  # Remove indicator code&name, and the last column which is all NAs
  select(-`Indicator Code`, -`Indicator Name`, -70) %>%
  rename(
    Entity = `Country Name`,
    Code = `Country Code`
  ) %>%
  pivot_longer(
    cols = 3:67,
    names_to = "Year",
    values_to = "Edu_Attainment"
  )

edu_attainment$Year <- as.numeric(edu_attainment$Year)

# Merge these two df's into the youth_not_in_eet

youth_not_in_eet <- youth_not_in_eet %>%
  left_join(. , edu_attainment, by = c("Code", "Year", "Entity")) %>%
  left_join(. , youth_unemployment, by  = c("Code", "Year", "Entity"))

write.csv(youth_not_in_eet, "Exported_data/youth_neet_cleaned.csv", row.names = FALSE)
```

Reference

World Bank (2025c). Educational attainment, at least completed upper secondary, population 25+, total (%) (cumulative) | Data. [online] World Bank Open Data. Available at: https://data.worldbank.org/indicator/SE.SEC.CUAT.UP.ZS [Accessed 1 Dec. 2025].

World Bank (2025a). Population, Total. [online] World Bank Open Data. Available at: https://data.worldbank.org/indicator/SP.POP.TOTL [Accessed 1 Dec. 2025].

World Bank (2025b). Unemployment, Youth Total (% of Total Labor Force Ages 15-24) (modeled ILO estimate) | Data. [online] World Bank Open Data. Available at: https://data.worldbank.org/indicator/SL.UEM.1524.ZS [Accessed 1 Dec. 2025].

LLM Usage

25/11/2025 asked ChatGPT "what would the definition of"least developed countries" mean in terms of gdp per capita (PPP)"

26/11/2025 asked ChatGPT "give me a list of all countries the UN considers LDCs between 2015 and 2021, with each country in quotation marks seperated by commas"

Limitation of the Data Sets

1. For the Youth NEET data set, there is no data for China, which may lead to a large error for conclusion drawn for Asia